The Classical Kicked Top

Lecture notes on the physics and on the `classical_kicked_top` code

Abstract

These notes explain the classical kicked top from the ground up: angular momentum as a dynamical variable, the sphere as a phase space, the kicked-top Hamiltonian and its stroboscopic map, energy, symmetries, fixed points and their stability, the transition from regular to chaotic motion, and Lyapunov exponents, with a short outlook on the quantum kicked top. The second part describes the numerical method and the structure of the Fortran/Python code, pointing each piece of code to the equation it implements. The only prerequisites are Hamiltonian mechanics (Hamilton's equations, Poisson brackets) and basic linear algebra. All phase-space figures were produced with the code described in part II.

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Part I
Physics

1 Angular momentum as a dynamical variable

1.1 Poisson brackets of angular momentum

Recall that for canonical coordinates q_a and momenta p_a the Poisson bracket of two phase-space functions is

$$\{A, B\} = \sum_a \left(\frac{\partial A}{\partial q_a}\frac{\partial B}{\partial p_a} - \frac{\partial A}{\partial p_a}\frac{\partial B}{\partial q_a} \right), \qquad \{q_a, p_b\} = \delta_{ab}, \tag{1}$$

where δ_{ab} is the Kronecker delta (1 if $a = b$, 0 otherwise), and that any observable $A(q, p, t)$ evolves as

$$\frac{dA}{dt} = \{A, H\} + \frac{\partial A}{\partial t}. \tag{2}$$

Take a particle with position $\boldsymbol{r}$ and momentum $\boldsymbol{p}$ and its angular momentum $\boldsymbol{J} = \boldsymbol{r} \times \boldsymbol{p}$, i.e. $J_i = \epsilon_{ijk} r_j p_k$. Here the indices i, j, k, m run over the three axes x, y, z (they are unrelated to the parameters j and k introduced in section 2), repeated indices are summed over, and ϵ_{ijk} is the Levi-Civita symbol: $\epsilon_{xyz} = \epsilon_{yzx} = \epsilon_{zxy} = 1$, $\epsilon_{xzy} = \epsilon_{zyx} = \epsilon_{yxz} = -1$, and $\epsilon_{ijk} = 0$ if two indices are equal. With it the cross product reads $(\boldsymbol{a} \times \boldsymbol{b})_i = \epsilon_{imk}\, a_m b_k$. A short calculation gives, for example,

$$\{J_x, J_y\} = \{yp_z - zp_y, zp_x - xp_z\} = y\,\{p_z, z\}\,p_x + p_y\,\{z, p_z\}\,x = -yp_x + xp_y = J_z,$$

and in general

$$\boxed{\{J_i, J_j\} = \epsilon_{ijk}\, J_k.} \tag{3}$$

The key idea of these notes is that (3) is *all* we need. We can forget $\boldsymbol{r}$ and $\boldsymbol{p}$ and treat the three components J_x, J_y, J_z themselves as the dynamical variables, with the bracket (3). This is how one describes the orientation of a rigid body's angular momentum, the total angular momentum of many atoms, or the classical limit of a *spin*. A spin is an angular momentum that has no $\boldsymbol{r} \times \boldsymbol{p}$ behind it, but whose quantum operators satisfy $[\hat{J}_i, \hat{J}_j] = i\hbar\, \epsilon_{ijk} \hat{J}_k$, the quantum version of (3) (the usual correspondence is $\{\cdot,\cdot\} \leftrightarrow [\cdot,\cdot]/i\hbar$). In this sentence only, a hat marks a quantum operator, $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ is the commutator, i the imaginary unit and $\hbar$ Planck's constant. Everywhere else in Part I a hat on a bold symbol, such as $\hat{\boldsymbol{y}}$, denotes a unit vector.

For any function $f(\boldsymbol{J})$, the chain rule and (3) give

$$\{J_i, f\} = \frac{\partial f}{\partial J_m}\{J_i, J_m\} = \epsilon_{imk}\, \frac{\partial f}{\partial J_m}\, J_k. \tag{4}$$

1.2 The length of $\boldsymbol{J}$ is conserved: the phase space is a sphere

Let $J^2 = J_x^2 + J_y^2 + J_z^2$. Using (4),

$$\{J_i, J^2\} = \epsilon_{imk}\, 2 J_m J_k = 0, \tag{5}$$

because ϵ_{imk} is antisymmetric in m, k while $J_m J_k$ is symmetric. Hence J^2 has a vanishing bracket with every function of $\boldsymbol{J}$ (it is a *Casimir* of the bracket), and in particular with every Hamiltonian $H(\boldsymbol{J}, t)$:

$$\frac{dJ^2}{dt} = \{J^2, H\} = 0. \tag{6}$$

Whatever the Hamiltonian, the vector $\boldsymbol{J}$ moves on a sphere of fixed radius, which we call j:

$$|\boldsymbol{J}| = j. \tag{7}$$

This sphere is the phase space of the system. It is two-dimensional, so the system has *one degree of freedom*, just like a particle on a line. The difference is that the phase space is compact (finite area) and curved.

1.3 Canonical coordinates on the sphere

Parametrise the sphere with the usual polar angle θ (measured from the z axis) and azimuth ϕ (fig. 1):

$$\boldsymbol{J} = j\,(\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta). \tag{8}$$

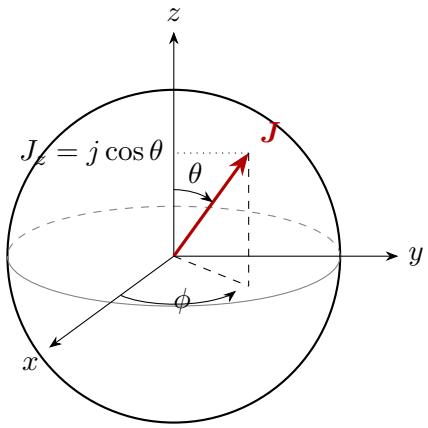


Figure 1: Coordinates on the phase-space sphere of radius j. The canonical pair is (ϕ, J_z).

Theorem. *The pair* $(q,p) = (\phi, J_z)$ *is canonical:* $\{\phi, J_z\} = 1$.

Proof. Write $J_x = R\cos\phi$, $J_y = R\sin\phi$ with $R = \sqrt{j^2 - J_z^2}$, so that $\partial R/\partial J_z = -J_z/R$, and compute brackets with $\{A,B\} = \partial_\phi A\,\partial_{J_z} B - \partial_{J_z} A\,\partial_\phi B$, where $\partial_\phi A \equiv \partial A/\partial\phi$ (the canonical bracket for $q=\phi$, $p=J_z$):

$$\{J_x, J_y\} = (-R\sin\phi)\Big(-\tfrac{J_z}{R}\sin\phi\Big) - \Big(-\tfrac{J_z}{R}\cos\phi\Big)(R\cos\phi) = J_z,$$
$$\{J_y, J_z\} = R\cos\phi = J_x, \qquad \{J_z, J_x\} = -(-R\sin\phi) = J_y.$$

This reproduces (3), so (ϕ, J_z) is a valid canonical pair. □

The phase-space (Liouville) area element is therefore $d\phi\,dJ_z$. The geometric area element of the sphere is $j^2\sin\theta\,d\theta\,d\phi$, and since $dJ_z = -j\sin\theta\,d\theta$,

$$j^2\sin\theta\,d\theta\,d\phi = j\ d\phi\,dJ_z. \tag{9}$$

Phase-space area and geometric area on the sphere are the same thing (up to the constant j). This is Archimedes' "hat-box" theorem: the band between two heights $z_1 < z_2$ has area proportional to $z_2 - z_1$. The total phase-space area is $4\pi j$. For this reason all phase-space plots in these notes use the axes $(\phi, \cos\theta)$ and not (ϕ,θ): equal areas in the plot are equal areas on the sphere.

1.4 Linear Hamiltonians generate rotations

Take $H = \boldsymbol{\omega}\cdot\boldsymbol{J}$ with a constant vector $\boldsymbol{\omega}$. With (2) and (4),

$$\frac{dJ_i}{dt} = \{J_i, \omega_m J_m\} = \epsilon_{imk}\,\omega_m J_k \quad\Longleftrightarrow\quad \frac{d\boldsymbol{J}}{dt} = \boldsymbol{\omega}\times\boldsymbol{J}. \tag{10}$$

This is *precession*. $\boldsymbol{J}$ rotates rigidly about the axis $\hat{\boldsymbol{\omega}} = \boldsymbol{\omega}/|\boldsymbol{\omega}|$ (the unit vector along $\boldsymbol{\omega}$) with angular velocity $|\boldsymbol{\omega}|$, counter-clockwise when looking down the axis (right-hand rule). After a time t the solution is $\boldsymbol{J}(t) = R_{\hat{\boldsymbol{\omega}}}(|\boldsymbol{\omega}|t)\,\boldsymbol{J}(0)$, where $R_{\hat{\boldsymbol{n}}}(\alpha)$ denotes the 3×3 matrix of a rotation by the angle α about the unit vector $\hat{\boldsymbol{n}}$ (counter-clockwise by the right-hand rule). We will need the rotations about y and z, $R_y \equiv R_{\hat{\boldsymbol{y}}}$ and $R_z \equiv R_{\hat{\boldsymbol{z}}}$, where $\hat{\boldsymbol{x}} = (1,0,0)$, $\hat{\boldsymbol{y}} = (0,1,0)$, $\hat{\boldsymbol{z}} = (0,0,1)$ are the unit vectors along the axes:

$$R_y(\alpha) = \begin{pmatrix} \cos\alpha & 0 & \sin\alpha \\ 0 & 1 & 0 \\ -\sin\alpha & 0 & \cos\alpha \end{pmatrix}, \qquad R_z(\alpha) = \begin{pmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{11}$$

Check: for $\boldsymbol{\omega} = \hat{\boldsymbol{y}}$, (10) gives $\dot{\boldsymbol{J}} = \hat{\boldsymbol{y}} \times \boldsymbol{J} = (J_z, 0, -J_x)$, where a dot denotes a time derivative, $\dot{\boldsymbol{J}} = d\boldsymbol{J}/dt$. Differentiating (11) at $\alpha = 0$ gives the same,

$$\frac{d}{d\alpha} R_y(\alpha)\Big|_{\alpha=0} \boldsymbol{J} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} \boldsymbol{J} = (J_z, 0, -J_x),$$

and in the same way $\frac{d}{d\alpha} R_z(\alpha)\boldsymbol{v} = \hat{\boldsymbol{z}} \times R_z(\alpha)\boldsymbol{v}$ for any vector $\boldsymbol{v}$. Rotation matrices are orthogonal, $R^{\mathsf{T}}R = I$ (^{T} is the transpose, I the identity matrix), so they preserve lengths, $|R\boldsymbol{v}| = |\boldsymbol{v}|$, and $R^{-1} = R^{\mathsf{T}}$. They also preserve cross products, $R(\boldsymbol{a} \times \boldsymbol{b}) = (R\boldsymbol{a}) \times (R\boldsymbol{b})$.

A Hamiltonian that is a function of $\boldsymbol{J}$ but not linear (for example J_z^2) still generates a rotation at each instant, but the axis and the rate depend on where $\boldsymbol{J}$ is. That nonlinearity is the source of chaos.

2 The kicked top

2.1 Hamiltonian

The kicked top is an angular momentum $\boldsymbol{J}$ with $|\boldsymbol{J}| = j$ and the time-periodic Hamiltonian

$$\boxed{H(t) = p\,J_y + \frac{k}{2j}\,J_z^2 \sum_{n=-\infty}^{\infty} \delta(t-n),} \tag{12}$$

with two real parameters:

- p, the angle of free precession about the y axis accumulated in one period (the period is set to 1);
- k, the strength of the *kick*, also called the *torsion* or *twist*.

Here $\delta(t)$ is the Dirac delta function: $\delta(t) = 0$ for $t \neq 0$, and $\int_a^b f(t)\,\delta(t-t_0)\,dt = f(t_0)$ for any smooth f and any interval $a < t_0 < b$. The sum $\sum_n \delta(t-n)$ (a *Dirac comb*) switches the second term on only at the integer times $t = n$, each time as an infinitely short, infinitely strong pulse. The first term is linear, so it generates a uniform rotation about y (section 1.4). The second term acts only at the instants $t = n$ (integer n). It is quadratic in J_z: as shown below, it rotates $\boldsymbol{J}$ about the z axis by an angle proportional to J_z. Points on the upper hemisphere are turned one way, points on the lower hemisphere the other way, and the equator not at all, like twisting a towel. The factor $1/j$ makes the twist angle depend only on the *direction* of $\boldsymbol{J}$ (see below).

Because H depends explicitly on time, (2) with $A = H$ gives $dH/dt = \{H, H\} + \partial H/\partial t = \partial H/\partial t \neq 0$ (the bracket of anything with itself vanishes), so energy is *not* conserved (section 4). With one degree of freedom and no conserved quantity, the motion can be chaotic. In contrast, a time-independent one-degree-of-freedom system is always integrable.

2.2 Scaled variables: only the direction matters

Since $|\boldsymbol{J}| = j$ is fixed, only the direction of $\boldsymbol{J}$ can change. Introduce the unit vector

$$\boldsymbol{s} = \frac{\boldsymbol{J}}{j} = (X, Y, Z) = (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta), \qquad |\boldsymbol{s}| = 1. \tag{13}$$

Its components are written $s_1 = X$, $s_2 = Y$, $s_3 = Z$ (or s_x, s_y, s_z) whenever index notation is convenient.

The scaled Hamiltonian. Insert $J_y = jY$ and $J_z = jZ$ into (12):

$$H = p\,(jY) + \frac{k}{2j}\,(jZ)^2 \sum_n \delta(t-n) = j\Big[\,pY + \frac{k}{2}\,Z^2 \sum_n \delta(t-n)\Big].$$

The whole Hamiltonian is j times a function of $\boldsymbol{s}$ alone. We write $H = j\,h$ with

$$h(\boldsymbol{s}, t) = p\,Y + \frac{k}{2}\,Z^2 \sum_n \delta(t-n). \tag{14}$$

The equation of motion for $\boldsymbol{s}$. Let ∇_s denote the gradient with respect to the three components of $\boldsymbol{s}$, treated as independent variables:

$$\nabla_s\, h = \Big(\frac{\partial h}{\partial X},\ \frac{\partial h}{\partial Y},\ \frac{\partial h}{\partial Z}\Big), \qquad \text{e.g.} \quad \nabla_s\big(pY\big) = (0, p, 0), \quad \nabla_s\Big(\frac{k}{2}Z^2\Big) = (0, 0, kZ).$$

Now derive $\dot{\boldsymbol{s}}$ in three steps.

1. $s_i = J_i/j$ has no explicit time dependence, so by (2)
$$\frac{ds_i}{dt} = \frac{1}{j}\,\frac{dJ_i}{dt} = \frac{1}{j}\,\{J_i, H\}.$$
2. By (4) with $f = H$,
$$\{J_i, H\} = \epsilon_{imk}\,\frac{\partial H}{\partial J_m}\,J_k.$$
3. $H(\boldsymbol{J}) = j\,h(\boldsymbol{J}/j)$, so by the chain rule $\dfrac{\partial H}{\partial J_m} = j\,\dfrac{\partial h}{\partial s_m}\,\dfrac{\partial s_m}{\partial J_m} = j\,\dfrac{\partial h}{\partial s_m}\,\dfrac{1}{j} = \dfrac{\partial h}{\partial s_m}$, and $J_k = j\,s_k$.

Putting the three together,

$$\frac{ds_i}{dt} = \frac{1}{j}\,\epsilon_{imk}\,\frac{\partial h}{\partial s_m}\,j\,s_k = \epsilon_{imk}\,\big(\nabla_s h\big)_m\,s_k,$$

and the right-hand side is the i-th component of the cross product (section 1). Hence

$$\boxed{\frac{d\boldsymbol{s}}{dt} = \nabla_s\, h \times \boldsymbol{s}.} \tag{15}$$

Three remarks.

- The radius j has dropped out. The classical dynamics of the direction $\boldsymbol{s}$ is the same for every j, so we can work on the unit sphere. The value of j only matters in the quantum theory, where $1/j$ plays the role of Planck's constant (section 9).
- (15) keeps $\boldsymbol{s}$ on the unit sphere: $\frac{d}{dt}|\boldsymbol{s}|^2 = 2\,\boldsymbol{s}\cdot\dot{\boldsymbol{s}} = 2\,\boldsymbol{s}\cdot(\nabla_s h \times \boldsymbol{s}) = 0$, because a cross product is perpendicular to each of its factors.
- Comparing with (10), (15) says that at each instant $\boldsymbol{s}$ precesses about the vector $\boldsymbol{\omega} = \nabla_s h$, which depends on $\boldsymbol{s}$ when h is not linear.

For the kicked top (14),

$$\nabla_s\, h = p\,\hat{\boldsymbol{y}} + kZ \sum_n \delta(t-n)\,\hat{\boldsymbol{z}}. \tag{16}$$

3 The stroboscopic map

Because the Hamiltonian is periodic and the kicks are instantaneous, it is natural to look at the system once per period, just after each kick. Write $t = n^+$ for the instant just after the kick at $t = n$ and $t = n^-$ for the instant just before it (the limits from above and from below). Let $\boldsymbol{s}_n = \boldsymbol{s}(n^+)$. Integrating (15) from $t = n^+$ to $t = (n+1)^+$ gives a map $\boldsymbol{s}_{n+1} = F(\boldsymbol{s}_n)$, the *stroboscopic* or *Floquet* map. It is derived in two steps: free precession from n^+ to $(n+1)^-$, then the kick from $(n+1)^-$ to $(n+1)^+$.

Step 1: free precession. For $n < t < n+1$ every $\delta(t-n)$ vanishes, so (16) is $\nabla_s h = p\,\hat{\boldsymbol{y}}$ and (15) becomes $\dot{\boldsymbol{s}} = p\,\hat{\boldsymbol{y}} \times \boldsymbol{s}$. With $\hat{\boldsymbol{y}} \times \boldsymbol{s} = (0,1,0) \times (X,Y,Z) = (Z,\ 0,\ -X)$ this is

$$\dot{X} = pZ, \qquad \dot{Y} = 0, \qquad \dot{Z} = -pX.$$

Y is constant. Differentiating the first equation and using the third gives $\ddot{X} = p\dot{Z} = -p^2 X$, a harmonic oscillator of frequency p. With $\tau = t - n$ the time since the kick, and initial values (X, Y, Z) at $\tau = 0$, the solution is

$$X(\tau) = X\cos p\tau + Z\sin p\tau, \qquad Z(\tau) = \frac{\dot{X}(\tau)}{p} = -X\sin p\tau + Z\cos p\tau$$

(check: $X(0) = X$, $Z(0) = Z$, and $\dot{Z} = -p\,X(\tau)$). At $\tau = 1$, just before the next kick,

$$X' = X\cos p + Z\sin p, \qquad Y' = Y, \qquad Z' = -X\sin p + Z\cos p, \tag{17}$$

which is $\boldsymbol{s}' = R_y(p)\,\boldsymbol{s}_n$ with R_y from (11): a rotation about y by the angle p.

Step 2: the kick. Integrate (15) over a short interval $[n+1-\varepsilon,\ n+1+\varepsilon]$ and let $\varepsilon \to 0$. The precession term $p\,\hat{\boldsymbol{y}} \times \boldsymbol{s}$ is bounded (its size is at most p), so its contribution to the change of $\boldsymbol{s}$ is at most $2\varepsilon p \to 0$. Only the δ term survives:

$$\dot{\boldsymbol{s}} = kZ\,\delta(t-n-1)\,\hat{\boldsymbol{z}} \times \boldsymbol{s}, \qquad \hat{\boldsymbol{z}} \times \boldsymbol{s} = (-Y,\ X,\ 0). \tag{18}$$

In components, $\dot{Z} = 0$, so Z does not change during the kick and keeps its value Z'. The other two are $\dot{X} = -kZ'\,\delta(t-n-1)\,Y$ and $\dot{Y} = kZ'\,\delta(t-n-1)\,X$. Introduce the angle accumulated since the start of the interval,

$$\alpha(t) = kZ' \int_{n+1-\varepsilon}^{t} \delta(t''-n-1)\,dt'',$$

which rises from $\alpha = 0$ before the kick to $\alpha = kZ'$ after it, with $d\alpha/dt = kZ'\,\delta(t-n-1)$. By the chain rule $\dot{X} = (dX/d\alpha)\,\dot{\alpha}$, so the equations become

$$\frac{dX}{d\alpha} = -Y, \qquad \frac{dY}{d\alpha} = X,$$

a rotation about z with unit rate in α: $X = X'\cos\alpha - Y'\sin\alpha$, $Y = X'\sin\alpha + Y'\cos\alpha$. At the end of the kick, $\alpha = kZ'$:

$$X'' = X'\cos(kZ') - Y'\sin(kZ'), \qquad Y'' = X'\sin(kZ') + Y'\cos(kZ'), \qquad Z'' = Z', \tag{19}$$

i.e. $\boldsymbol{s}'' = R_z(kZ')\,\boldsymbol{s}'$. This is a *torsion*: a rotation about z whose angle depends on the height Z. (Replacing δ by any narrow pulse of unit area gives the same result, since only the total angle kZ' enters.)

The map. Together:

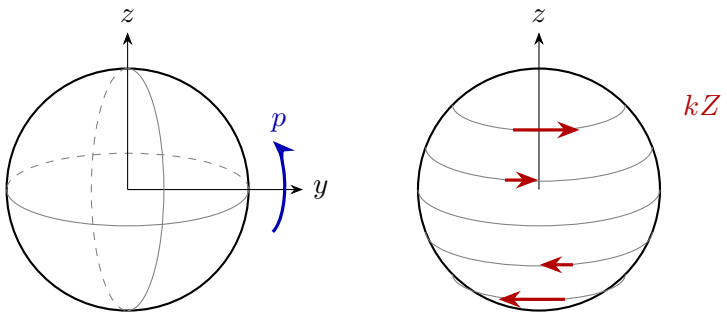
$$\boxed{\boldsymbol{s}_{n+1} = F(\boldsymbol{s}_n) = R_z\big(kZ'\big)\, R_y(p)\, \boldsymbol{s}_n, \qquad Z' = \big(R_y(p)\boldsymbol{s}_n\big)_z.} \tag{20}$$

Figure 2 illustrates the two steps. Two properties follow directly.

- Both steps are rotations, which preserve lengths (section 1.4), so $|\boldsymbol{s}_n| = 1$ holds exactly.
- The map is invertible. The torsion does not change Z, so the z component of $\boldsymbol{s}_{n+1}$ is Z' itself. Undoing the torsion and then the rotation gives

  $$F^{-1}(\boldsymbol{s}) = R_y(-p)\, R_z(-kZ)\, \boldsymbol{s},$$

  where Z is the z component of the argument $\boldsymbol{s}$.



(a) rotation about y by p (b) torsion about z by kZ

Figure 2: The two steps of one period of the kicked top, eq. (20). (a) A rigid rotation about y. (b) A rotation about z whose angle kZ grows with height: the upper and lower hemispheres are twisted in opposite directions and the equator stays put.

3.1 The map preserves phase-space area

If a map sends coordinates (q, P) to (q'', P''), a small area element is multiplied by $|\det D|$, where D is the *Jacobian matrix* of partial derivatives

$$D = \begin{pmatrix} \partial q''/\partial q & \partial q''/\partial P \\ \partial P''/\partial q & \partial P''/\partial P \end{pmatrix}$$

and det is its determinant. We use the canonical coordinates (ϕ, Z) of section 1.3.

The torsion. With $X = \sqrt{1-Z^2}\cos\phi$ and $Y = \sqrt{1-Z^2}\sin\phi$ (from (13)), the rotation (19) by the angle α reads, using the addition formulas for cos and sin,

$$X'' = \sqrt{1-Z'^2}\,\cos(\phi' + \alpha), \qquad Y'' = \sqrt{1-Z'^2}\,\sin(\phi' + \alpha), \qquad Z'' = Z'.$$

So a rotation about z only shifts ϕ by α. With $\alpha = kZ'$,

$$\phi'' = \phi' + kZ', \qquad Z'' = Z', \qquad D_{\text{torsion}} = \begin{pmatrix} \partial\phi''/\partial\phi' & \partial\phi''/\partial Z' \\ \partial Z''/\partial\phi' & \partial Z''/\partial Z' \end{pmatrix} = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}, \tag{21}$$

and $\det D_{\text{torsion}} = 1 \cdot 1 - k \cdot 0 = 1$. A map of this form, where one coordinate is shifted by an amount that grows with the other, is called a *twist map*.

The rotation. $R_y(p)$ moves the sphere rigidly, so it preserves geometric area on the sphere. By (9) geometric area and phase-space area $d\phi\, dZ$ are the same, so $|\det D_{\text{rotation}}| = 1$. (The same follows from Liouville's theorem, since $R_y(p)$ is the flow of the Hamiltonian pY.)

The full map. By the chain rule the Jacobian of a composition is the product of the Jacobians, and $\det(AB) = \det A \det B$. Hence

$$\det DF = \det D_{\text{torsion}} \det D_{\text{rotation}} = 1 \quad \text{(in the coordinates } (\phi, Z)\text{)}. \tag{22}$$

(The coordinates (ϕ, Z) fail at the poles $Z = \pm 1$, where ϕ is undefined, but that is a property of the coordinates and not of the map.) Two consequences will be used repeatedly. First, there are no attractors: a region of phase space can be stretched and folded but never shrunk. Second, linear stability is decided by a single number, the trace of the Jacobian (section 6).

4 Energy

The code evaluates, at every recorded point $\boldsymbol{s}_n$, the two terms of the scaled Hamiltonian (14) without the δ comb, and their sum:

$$E_{\text{rot}} = pY, \qquad E_{\text{kick}} = \frac{k}{2}Z^2, \qquad E = E_{\text{rot}} + E_{\text{kick}} = pY + \frac{k}{2}Z^2. \tag{23}$$

For $p, k \geq 0$, $|Y| \leq 1$ and $0 \leq Z^2 \leq 1$ give $-p \leq E \leq p + k/2$. The minimum $E = -p$ is reached at $-\hat{\boldsymbol{y}}$, where $Y = -1$ and $Z = 0$.

Why E is not conserved. Follow E through one period with (17) and (19).

- During the precession Y is constant, so E_{rot} is unchanged, but $Z \to Z' = -X \sin p + Z \cos p$ changes E_{kick}.
- During the kick Z is constant, so E_{kick} is unchanged, but $Y \to Y'' = X' \sin(kZ') + Y' \cos(kZ')$ changes E_{rot}.

Over one period

$$E_{n+1} - E_n = p\,(Y'' - Y) + \frac{k}{2}(Z'^2 - Z^2),$$

which is not zero in general. This is the explicit form of $dH/dt = \partial H/\partial t \neq 0$ (section 2). There are, however, two reasons why E is a useful quantity.

The near-continuous limit. Suppose both p and k are small. Expand both steps to first order. A rotation by a small angle α about $\hat{\boldsymbol{n}}$ is $R_{\hat{\boldsymbol{n}}}(\alpha)\boldsymbol{s} = \boldsymbol{s} + \alpha\, \hat{\boldsymbol{n}} \times \boldsymbol{s} + O(\alpha^2)$ (the derivative of the rotation at $\alpha = 0$, as checked in section 1.4). So

$$\begin{aligned} \boldsymbol{s}' &= \boldsymbol{s} + p\, \hat{\boldsymbol{y}} \times \boldsymbol{s} + O(p^2), \\ \boldsymbol{s}_{n+1} &= \boldsymbol{s}' + kZ'\, \hat{\boldsymbol{z}} \times \boldsymbol{s}' + O(k^2) = \boldsymbol{s} + \big(p\, \hat{\boldsymbol{y}} + kZ\, \hat{\boldsymbol{z}}\big) \times \boldsymbol{s} + O(p^2, pk, k^2), \end{aligned}$$

where we used $Z' = Z + O(p)$ and $\boldsymbol{s}' = \boldsymbol{s} + O(p)$. Now $p\, \hat{\boldsymbol{y}} + kZ\, \hat{\boldsymbol{z}} = (0,\, p,\, kZ) = \nabla_s\, h_{\text{eff}}$ with the *effective Hamiltonian*

$$h_{\text{eff}} = pY + \frac{k}{2}Z^2 = E. \tag{24}$$

So $\boldsymbol{s}_{n+1} = \boldsymbol{s}_n + \nabla_s h_{\text{eff}} \times \boldsymbol{s}_n + O(p^2, pk, k^2)$: one period is, to first order, a step of length 1 of the continuous motion $\dot{\boldsymbol{s}} = \nabla_s h_{\text{eff}} \times \boldsymbol{s}$. (The Baker–Campbell–Hausdorff formula makes this

systematic: the exact time-1 flow of h_{eff} and the map F differ only at order pk.) The flow of a time-independent Hamiltonian conserves it:

$$\frac{dh_{\text{eff}}}{dt} = \nabla_s h_{\text{eff}} \cdot \dot{\boldsymbol{s}} = \nabla_s h_{\text{eff}} \cdot \big(\nabla_s h_{\text{eff}} \times \boldsymbol{s}\big) = 0.$$

So in this limit E is nearly conserved and orbits follow the level curves of E. Figure 3 shows this for $p = 0.1$, $k = 0.2$: every orbit traces a curve of (almost) constant colour.

Fixed points of the effective flow. The flow stops where $\dot{\boldsymbol{s}} = \nabla_s h_{\text{eff}} \times \boldsymbol{s} = 0$, i.e. where $\nabla_s h_{\text{eff}}$ is parallel to $\boldsymbol{s}$: $\nabla_s h_{\text{eff}} = \mu\, \boldsymbol{s}$ for some number μ. (This is also the Lagrange condition for the extrema of h_{eff} on the sphere, with μ the Lagrange multiplier.) In components,

$$0 = \mu X, \qquad p = \mu Y, \qquad kZ = \mu Z.$$

For $p \neq 0$ the second equation forces $\mu \neq 0$, so the first gives $X = 0$. The third has two solutions.

- $Z = 0$: then $Y = \pm 1$, the poles $\pm\hat{\boldsymbol{y}}$.
- $\mu = k$: then $Y = p/k$ and $Z = \pm\sqrt{1 - p^2/k^2}$, which exists only if $k \geq p$.

For $p = 0.1$, $k = 0.2$ the second pair is at $Y = 0.5$, $X = 0$, i.e. $\phi = \pi/2$, $Z = \pm 0.87$: the two centres visible in fig. 3.

To see which points are centres and which are saddles, expand h_{eff} near $\sigma\hat{\boldsymbol{y}}$ ($\sigma = \pm 1$) with small $x = X$, $z = Z$. From $Y = \sigma\sqrt{1 - x^2 - z^2} \approx \sigma\big(1 - \frac{1}{2}(x^2 + z^2)\big)$,

$$h_{\text{eff}} \approx \sigma p - \frac{\sigma p}{2}\, x^2 + \frac{k - \sigma p}{2}\, z^2.$$

At $-\hat{\boldsymbol{y}}$ ($\sigma = -1$) both coefficients are positive, so h_{eff} has a minimum there. Its level curves are small closed curves around the point, so it is a *centre*. At $+\hat{\boldsymbol{y}}$ the coefficients are $-p/2$ and $(k - p)/2$. For $k < p$ both are negative (a maximum, a centre). For $k > p$ they have opposite signs, the level curves are hyperbolas, and $+\hat{\boldsymbol{y}}$ is a *saddle*: orbits approach it along one direction and leave along another. This is the case in fig. 3 ($\phi = \pi/2$, $Z = 0$), where the curve through the saddle (the *separatrix*) is clearly visible.

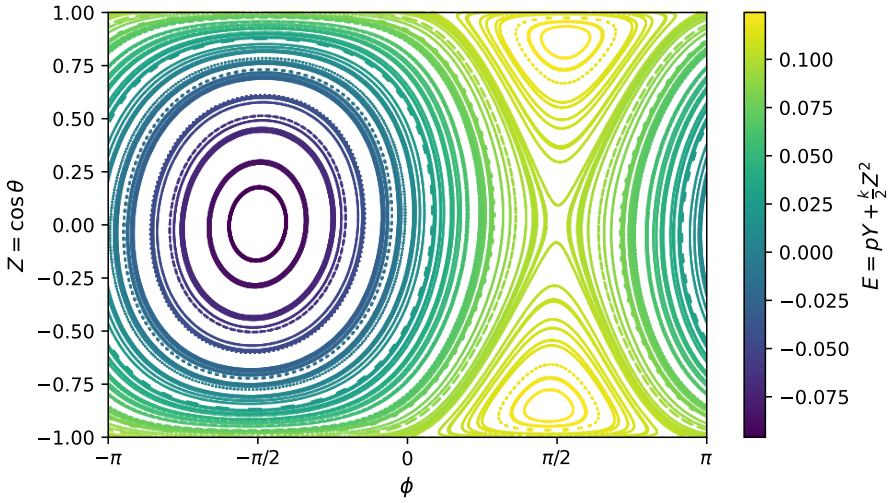


Figure 3: Phase portrait for small parameters $p = 0.1$, $k = 0.2$ (12×12 grid of initial conditions, 1000 kicks each), coloured by E of eq. (23). Orbits lie on level curves of E, as predicted by eq. (24).

Away from that limit. For $p = \pi/2$ the near-continuous picture does not apply and E fluctuates from kick to kick (fig. 4). On a regular orbit (left) the fluctuations are quasi-periodic and stay inside a narrow band. On a chaotic orbit (right) they are erratic, and E wanders over a large part of its range $[-p,\ p + k/2]$. The behaviour of E_n is thus a simple diagnostic of regular versus chaotic motion.

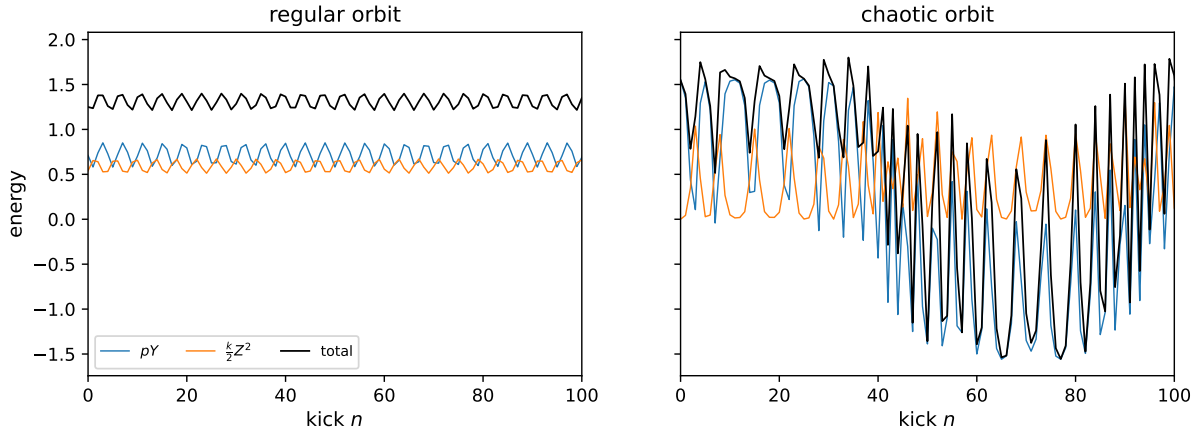


Figure 4: The energy terms (23) along a regular orbit (left, started inside the island around the fixed point $\boldsymbol{s}_+$ of section 6.3) and along a chaotic orbit (right, started next to the unstable pole $+\hat{\boldsymbol{y}}$), for $p = \pi/2$, $k = 3$.

5 Integrable limits and symmetry

Integrable limits. A map is called *integrable* if there is a conserved function $I(\boldsymbol{s})$, $I(F(\boldsymbol{s})) = I(\boldsymbol{s})$. Each orbit then stays on one level curve $I = \text{const}$, an *invariant curve*.

- $k = 0$: the torsion is $R_z(0)$, the identity, and $F = R_y(p)$. Y is conserved. Every orbit lies on a circle $Y = \text{const}$ and advances by the angle p per kick. If $p/2\pi = r/q$ with integers r, q, then $R_y(p)^q = R_y(2\pi r)$ is the identity and every orbit is periodic with period q (for $p = \pi/2$, $q = 4$).
- $p = 0$: F is the pure twist (21) and Z is conserved. Each circle of latitude rotates rigidly by its own angle, $\phi_n = \phi_0 + n\,kZ$.

When both p and k are non-zero there is in general no conserved quantity.

A symmetry. Let $S = R_y(\pi)$. From (11) with $\cos\pi = -1$, $\sin\pi = 0$,

$$S = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad (X, Y, Z) \mapsto (-X,\ Y,\ -Z), \qquad S^{-1} = S.$$

We show that $F(S\boldsymbol{s}) = S\,F(\boldsymbol{s})$.

1. Rotations about the same axis commute, $R_y(\pi)R_y(p) = R_y(\pi + p) = R_y(p)R_y(\pi)$. So $R_y(p)\,S\boldsymbol{s} = S\,\boldsymbol{s}'$ with $\boldsymbol{s}' = R_y(p)\boldsymbol{s}$.
2. S flips the sign of the z component, so the torsion angle for $S\boldsymbol{s}'$ is $-kZ'$. Hence $F(S\boldsymbol{s}) = R_z(-kZ')\,S\,\boldsymbol{s}'$.

3. Multiply out $SR_z(\alpha)S$ with (11): each entry $(R_z)_{ab}$ gets the factor $S_{aa}S_{bb}$, which flips the sign of the two $\sin\alpha$ entries and nothing else. So $SR_z(\alpha)S = R_z(-\alpha)$, or $R_z(-\alpha)\,S = S\,R_z(\alpha)$.

4. With $\alpha = kZ'$: $F(S\boldsymbol{s}) = R_z(-kZ')\,S\,\boldsymbol{s}' = S\,R_z(kZ')\,\boldsymbol{s}' = S\,F(\boldsymbol{s})$.

The two sign changes (of the axis and of the height) cancel:

$$F(S\boldsymbol{s}) = S\,F(\boldsymbol{s}). \tag{25}$$

If $\boldsymbol{s}_0, \boldsymbol{s}_1, \ldots$ is an orbit, then $S\boldsymbol{s}_{n+1} = SF(\boldsymbol{s}_n) = F(S\boldsymbol{s}_n)$, so $S\boldsymbol{s}_0, S\boldsymbol{s}_1, \ldots$ is also an orbit. In the coordinates (ϕ, Z), S is $(\phi, Z) \mapsto (\pi - \phi, -Z)$, since $\cos(\pi - \phi) = -\cos\phi$ and $\sin(\pi - \phi) = \sin\phi$. This is a point reflection about $(\pi/2, 0)$ (with ϕ taken modulo 2π), and every phase portrait in these notes is symmetric under it. Orbits either come in symmetric pairs or are symmetric themselves.

6 Fixed points, periodic orbits and their stability

6.1 Linear stability of an area-preserving map

A point $\boldsymbol{s}^*$ with $F(\boldsymbol{s}^*) = \boldsymbol{s}^*$ is a *fixed point*. A point with $F^m(\boldsymbol{s}^*) = \boldsymbol{s}^*$, where $F^m = F \circ \cdots \circ F$ is F applied m times and m is the smallest such number, lies on a *periodic orbit* of period m.

Linearisation. To see what happens near $\boldsymbol{s}^*$, write $\boldsymbol{s} = \boldsymbol{s}^* + \delta\boldsymbol{s}$, where $\delta\boldsymbol{s}$ (here δ means "a small change of", not the Dirac function) is small. Since $|\boldsymbol{s}| = |\boldsymbol{s}^*| = 1$, to first order $\boldsymbol{s}^* \cdot \delta\boldsymbol{s} = 0$: $\delta\boldsymbol{s}$ lies in the *tangent plane* of the sphere at $\boldsymbol{s}^*$. Choose two orthogonal unit vectors $\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2$ spanning that plane and write $\delta\boldsymbol{s} = a_1\hat{\boldsymbol{e}}_1 + a_2\hat{\boldsymbol{e}}_2$. A Taylor expansion of F to first order gives

$$F(\boldsymbol{s}^* + \delta\boldsymbol{s}) = \boldsymbol{s}^* + M\,\delta\boldsymbol{s} + O(\delta\boldsymbol{s}^2) \qquad \Longrightarrow \qquad \delta\boldsymbol{s}_{n+1} = M\,\delta\boldsymbol{s}_n, \tag{26}$$

where M is the 2×2 Jacobian matrix $DF(\boldsymbol{s}^*)$ in the basis $(\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2)$: its entry M_{ab} is the $\hat{\boldsymbol{e}}_a$ component of the change of F when $\boldsymbol{s}^*$ is moved by a unit amount along $\hat{\boldsymbol{e}}_b$. For a period-m orbit $\boldsymbol{s}_0^* \to \boldsymbol{s}_1^* \to \cdots \to \boldsymbol{s}_m^* = \boldsymbol{s}_0^*$ the chain rule gives the product of the Jacobians along the orbit, $M^{(m)} = DF(\boldsymbol{s}_{m-1}^*) \cdots DF(\boldsymbol{s}_1^*)\,DF(\boldsymbol{s}_0^*)$.

The trace decides. The trace $\operatorname{Tr} M = M_{11} + M_{22}$ and the determinant $\det M$ do not depend on the basis: a change of basis replaces M by PMP^{-1}, and $\operatorname{Tr}(PMP^{-1}) = \operatorname{Tr} M$, $\det(PMP^{-1}) = \det M$. Area preservation (section 3.1) gives $\det M = 1$. The eigenvalues λ of M ($M\boldsymbol{v} = \lambda\boldsymbol{v}$ for some vector $\boldsymbol{v} \neq 0$) solve

$$\det(M - \lambda I) = \lambda^2 - (\operatorname{Tr} M)\,\lambda + \det M = \lambda^2 - (\operatorname{Tr} M)\,\lambda + 1 = 0, \qquad \lambda_{1,2} = \frac{\operatorname{Tr} M \pm \sqrt{(\operatorname{Tr} M)^2 - 4}}{2},$$

with $\lambda_1\lambda_2 = 1$. Three cases:

- $|\operatorname{Tr} M| < 2$: **elliptic** (stable). The square root is imaginary, $\lambda_{1,2} = e^{\pm i\alpha}$ with $\lambda_1 + \lambda_2 = 2\cos\alpha = \operatorname{Tr} M$. In a suitable basis M acts as a rotation by α, so nearby points circle around $\boldsymbol{s}^*$ by about α per kick on small ellipses and stay close. In the phase portrait an elliptic point sits at the centre of a family of closed curves, an *island*.

- $|\operatorname{Tr} M| > 2$: **hyperbolic** (unstable). Two real eigenvalues λ_u and $1/\lambda_u$ with $|\lambda_u| > 1$. A displacement along the eigenvector of λ_u grows like $|\lambda_u|^n$, one along the other eigenvector shrinks like $|\lambda_u|^{-n}$. Hyperbolic points are the seeds of chaos.

- $|\operatorname{Tr} M| = 2$: **parabolic**, $\lambda_1 = \lambda_2 = \pm 1$. This is the boundary case at which **bifurcations** happen. At $\operatorname{Tr} M = +2$ ($\lambda = 1$) new fixed points can branch off. At $\operatorname{Tr} M = -2$ ($\lambda = -1$) a displacement flips sign every kick and comes back after two, and the fixed point typically gives birth to a period-2 orbit (*period doubling*).

6.2 The poles $\pm\hat{\boldsymbol{y}}$

The two points $\boldsymbol{s} = \sigma\hat{\boldsymbol{y}}$, $\sigma = \pm 1$, are fixed for all p and k. $R_y(p)$ leaves them alone, because they lie on its axis. Then $Z' = 0$, so the torsion angle kZ' is zero and the torsion does nothing.

Near $\sigma\hat{\boldsymbol{y}}$ write $\boldsymbol{s} = (x,\ \sigma\sqrt{1-x^2-z^2},\ z) \approx (x,\ \sigma,\ z)$ with small x, z. The tangent plane is spanned by $\hat{\boldsymbol{x}}, \hat{\boldsymbol{z}}$, and (x, z) are the coordinates in it.

- The rotation (17) acts on (x, z) as $x' = x\cos p + z\sin p$, $z' = -x\sin p + z\cos p$.
- The torsion (19) with $X' = x'$, $Y' \approx \sigma$ and the small angle kz' ($\cos kz' \approx 1$, $\sin kz' \approx kz'$) gives $x'' \approx x' - \sigma k z'$, $z'' = z'$, dropping products of small quantities.

Hence

$$M_\sigma = \begin{pmatrix} 1 & -\sigma k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \cos p & \sin p \\ -\sin p & \cos p \end{pmatrix} = \begin{pmatrix} \cos p + \sigma k \sin p & \sin p - \sigma k \cos p \\ -\sin p & \cos p \end{pmatrix}, \tag{27}$$

with $\det M_\sigma = (\cos p + \sigma k \sin p)\cos p + \sin p\,(\sin p - \sigma k \cos p) = \cos^2 p + \sin^2 p = 1$ and

$$\boxed{\operatorname{Tr} M_\sigma = 2\cos p + \sigma\, k \sin p.} \tag{28}$$

For $p = \pi/2$ ($\cos p = 0$, $\sin p = 1$) this is $\operatorname{Tr} M_\pm = \pm k$. Both poles are stable for $k < 2$, and both become unstable at $k = 2$: $+\hat{\boldsymbol{y}}$ through $\operatorname{Tr} = +2$ and $-\hat{\boldsymbol{y}}$ through $\operatorname{Tr} = -2$. For small p, $\cos p \approx 1 - p^2/2$ and $\sin p \approx p$ give $\operatorname{Tr} M_+ \approx 2 - p^2 + kp$, which exceeds 2 when $k > p$: the same threshold at which $+\hat{\boldsymbol{y}}$ became a saddle of h_{eff} in section 4.

6.3 The case $p = \pi/2$

From now on we use the standard choice $p = \pi/2$. With $\cos p = 0$, $\sin p = 1$, (17) becomes $(X', Y', Z') = (Z,\ Y,\ -X)$. The torsion angle is $kZ' = -kX$, and with $\cos(-kX) = \cos kX$, $\sin(-kX) = -\sin kX$, (19) gives

$$X_{n+1} = Z_n \cos(kX_n) + Y_n \sin(kX_n), \quad Y_{n+1} = -Z_n \sin(kX_n) + Y_n \cos(kX_n), \quad Z_{n+1} = -X_n. \tag{29}$$

The fixed points born at $k = 2$. A fixed point needs $Z_{n+1} = Z_n$, i.e. $Z = -X$. Write $a = kX$ and insert $Z = -X$ into the first two equations of (29), using $1 + \cos a = 2\cos^2\frac{a}{2}$, $1 - \cos a = 2\sin^2\frac{a}{2}$ and $\sin a = 2\sin\frac{a}{2}\cos\frac{a}{2}$:

$$\begin{aligned} X = -X\cos a + Y\sin a &\implies X(1+\cos a) = Y\sin a \implies X\cos\tfrac{a}{2} = Y\sin\tfrac{a}{2}, \\ Y = X\sin a + Y\cos a &\implies Y(1-\cos a) = X\sin a \implies Y\sin\tfrac{a}{2} = X\cos\tfrac{a}{2}. \end{aligned}$$

Both give the same condition $Y = X\cot(kX/2)$. ($X = 0$ gives $Z = 0$, $Y = \pm 1$: the poles again.) The normalisation $X^2 + Y^2 + Z^2 = 1$ then gives

$$g(X) \equiv 2X^2 + X^2\cot^2\Big(\frac{kX}{2}\Big) = 1, \qquad \boldsymbol{s}_\pm = \big(\pm X,\ X\cot(kX/2),\ \mp X\big). \tag{30}$$

$X\cot(kX/2)$ is even in X, so the solutions come in pairs $\pm X$ with the same Y. For small u, $\cot u \approx 1/u$, so $X\cot(kX/2) \to 2/k$ and $g(X) \to 4/k^2$ as $X \to 0$.

- $k < 2$: $g(0) = 4/k^2 > 1$, and a numerical scan confirms that $g(X) = 1$ has no solution.
- $k = 2$: $g(0) = 1$, the solution sits at $X = 0$, $Y = 2/k = 1$, i.e. at the pole $+\hat{\boldsymbol{y}}$.
- $k > 2$: a pair $\boldsymbol{s}_\pm$ with small X moves away from $+\hat{\boldsymbol{y}}$.

So $\boldsymbol{s}_\pm$ branch off the pole $+\hat{\boldsymbol{y}}$ exactly at $k = 2$, where the pole's trace passes $+2$: a *pitchfork bifurcation* (one fixed point becomes three). The two points are mapped onto each other by S of (25): $S(X, Y, -X) = (-X, Y, X)$. Solving (30) numerically and computing the trace of the Jacobian of (29) in the tangent plane:

k	$\boldsymbol{s}_+ = (X, Y, Z)$	(ϕ, Z)	$\operatorname{Tr} M$
2.5	(0.514, 0.687, −0.514)	(0.928, −0.514)	1.00
3.0	(0.629, 0.456, −0.629)	(0.627, −0.629)	0.06
3.5	(0.681, 0.272, −0.681)	(0.380, −0.681)	−0.77
4.0	(0.702, 0.118, −0.702)	(0.167, −0.702)	−1.47

Period doubling of $\boldsymbol{s}_\pm$ at $k = \pi\sqrt{2}$. The trace decreases with k and reaches -2 at $k = \pi\sqrt{2} \approx 4.443$. This can be checked exactly. At $k = \pi\sqrt{2}$ and $X = 1/\sqrt{2}$, $kX/2 = \pi/2$ and $\cot(\pi/2) = 0$, so $Y = 0$ and $g = 2 \cdot \frac{1}{2} = 1$: (30) is solved by $\boldsymbol{s}_+ = (1, 0, -1)/\sqrt{2}$. Its tangent plane is spanned by $\hat{\boldsymbol{u}} = \hat{\boldsymbol{y}}$ and $\hat{\boldsymbol{w}} = (\hat{\boldsymbol{x}} + \hat{\boldsymbol{z}})/\sqrt{2}$. Write $\boldsymbol{s} = \boldsymbol{s}_+ + a\,\hat{\boldsymbol{u}} + b\,\hat{\boldsymbol{w}}$:

$$X = \frac{1+b}{\sqrt{2}}, \qquad Y = a, \qquad Z = \frac{-1+b}{\sqrt{2}}, \qquad kX = \pi(1+b),$$

so to first order $\cos kX = -\cos \pi b \approx -1$ and $\sin kX = -\sin \pi b \approx -\pi b$. Inserting into (29) and dropping products of a, b:

$$X_{n+1} \approx \frac{1-b}{\sqrt{2}}, \qquad Y_{n+1} \approx -\frac{\pi}{\sqrt{2}}\, b - a, \qquad Z_{n+1} = -\frac{1+b}{\sqrt{2}},$$

i.e. $\boldsymbol{s}_{n+1} = \boldsymbol{s}_+ + a'\hat{\boldsymbol{u}} + b'\hat{\boldsymbol{w}}$ with $a' = -a - \frac{\pi}{\sqrt{2}} b$, $b' = -b$. Hence

$$M = \begin{pmatrix} -1 & -\pi/\sqrt{2} \\ 0 & -1 \end{pmatrix}, \qquad \det M = 1, \qquad \operatorname{Tr} M = -2.$$

Beyond $k \approx 4.44$ the fixed points $\boldsymbol{s}_\pm$ are unstable (period doubling), and their islands disappear from the phase portraits (fig. 5, $k = 4.5$).

A period-2 orbit born at $k = 2$. At $k = 2$ the pole $-\hat{\boldsymbol{y}}$ passes $\operatorname{Tr} = -2$ and, by the period doubling scenario of section 6, emits a period-2 orbit: two points $\boldsymbol{s}_A, \boldsymbol{s}_B$ with $F(\boldsymbol{s}_A) = \boldsymbol{s}_B$, $F(\boldsymbol{s}_B) = \boldsymbol{s}_A$. For $k = 3$ a numerical search gives $(\phi, Z) = (-0.627,\ 0.629)$ and $(-2.515,\ -0.629)$, with $\operatorname{Tr} M^{(2)} = \operatorname{Tr}\left[DF(\boldsymbol{s}_B)\, DF(\boldsymbol{s}_A)\right] = -1.93$ (stable, close to its own period doubling).

A period-4 orbit for every k. Follow $\hat{\boldsymbol{x}}$ under (29), using $R_y(\pi/2) : (X, Y, Z) \mapsto (Z, Y, -X)$:

$$\hat{\boldsymbol{x}} \xrightarrow{R_y} -\hat{\boldsymbol{z}} \xrightarrow{\text{torsion}} -\hat{\boldsymbol{z}}, \qquad -\hat{\boldsymbol{z}} \xrightarrow{R_y} -\hat{\boldsymbol{x}} \xrightarrow{\text{torsion}} -\hat{\boldsymbol{x}},$$
$$-\hat{\boldsymbol{x}} \xrightarrow{R_y} \hat{\boldsymbol{z}} \xrightarrow{\text{torsion}} \hat{\boldsymbol{z}}, \qquad \hat{\boldsymbol{z}} \xrightarrow{R_y} \hat{\boldsymbol{x}} \xrightarrow{\text{torsion}} \hat{\boldsymbol{x}}.$$

After each rotation the point lies either on the z axis (which a rotation about z does not move) or on the equator (where the torsion angle is zero). So $\hat{\boldsymbol{x}} \to -\hat{\boldsymbol{z}} \to -\hat{\boldsymbol{x}} \to \hat{\boldsymbol{z}} \to \hat{\boldsymbol{x}}$ is a period-4 orbit for every k.

Its stability needs the linearised torsion. Differentiate $T(\boldsymbol{s}) = R_z(kZ)\,\boldsymbol{s}$, using the derivative of R_z from section 1.4:

$$DT(\boldsymbol{s})\,\delta\boldsymbol{s} = R_z(kZ)\,\delta\boldsymbol{s} + k\,\delta Z\ \hat{\boldsymbol{z}} \times R_z(kZ)\,\boldsymbol{s}.$$

At the poles $\pm\hat{\boldsymbol{z}}$ the second term vanishes ($\hat{\boldsymbol{z}} \times \hat{\boldsymbol{z}} = 0$) and the torsion acts on the tangent plane as the rotation $R_z(\pm k)$. On the equator at $\pm\hat{\boldsymbol{x}}$, $R_z(0) = I$ and $\hat{\boldsymbol{z}} \times (\pm\hat{\boldsymbol{x}}) = \pm\hat{\boldsymbol{y}}$, so

$DT\,\delta\boldsymbol{s} = \delta\boldsymbol{s} \pm k\,\delta Z\,\hat{\boldsymbol{y}}$, a shear. Use the tangent bases $(\hat{\boldsymbol{y}}, \hat{\boldsymbol{z}})$ at $\pm\hat{\boldsymbol{x}}$ and $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}})$ at $\pm\hat{\boldsymbol{z}}$. Following a displacement through $R_y(\pi/2)$ ($\hat{\boldsymbol{y}} \to \hat{\boldsymbol{y}}$, $\hat{\boldsymbol{z}} \to \hat{\boldsymbol{x}}$, $\hat{\boldsymbol{x}} \to -\hat{\boldsymbol{z}}$) and then the torsion gives the four 2×2 Jacobians

$$\hat{\boldsymbol{x}} \to -\hat{\boldsymbol{z}}:\ D_1 = \begin{pmatrix} \sin k & \cos k \\ \cos k & -\sin k \end{pmatrix}, \qquad -\hat{\boldsymbol{z}} \to -\hat{\boldsymbol{x}}:\ D_2 = \begin{pmatrix} k & 1 \\ -1 & 0 \end{pmatrix},$$
$$-\hat{\boldsymbol{x}} \to \hat{\boldsymbol{z}}:\ D_3 = \begin{pmatrix} -\sin k & \cos k \\ \cos k & \sin k \end{pmatrix}, \qquad \hat{\boldsymbol{z}} \to \hat{\boldsymbol{x}}:\ D_4 = \begin{pmatrix} -k & 1 \\ -1 & 0 \end{pmatrix}.$$

(The bases are not all oriented the same way, so $\det D_1 = \det D_3 = -1$. Only the product is meaningful, and it has determinant 1.) Multiplying out $M^{(4)} = D_4 D_3 D_2 D_1$ and simplifying with $\cos^2 k - \sin^2 k = \cos 2k$, $2\sin k \cos k = \sin 2k$,

$$\boxed{\operatorname{Tr} M^{(4)} = 2\cos 2k + 2k \sin 2k + k^2 \sin^2 k.} \tag{31}$$

For small k, $\operatorname{Tr} M^{(4)} = 2 - k^4/3 + O(k^6)$, just below 2 (marginally stable). The trace depends on k in a non-monotonic way:

k	0.5	1.5	2.5	3.0	4.5	6.0
$\operatorname{Tr} M^{(4)}$	1.98	0.68	−1.99	0.42	21.2	−1.94

These values agree with a direct numerical evaluation of the Jacobians. The orbit is stable at $k = 3$ and at $k = 6$: its islands are visible around $(\phi, Z) = (0, 0)$, $(\pm\pi, 0)$ and near the poles $Z = \pm 1$ in fig. 5. At $k = 4.5$ it is unstable.

7 From regular to chaotic motion

7.1 Regular motion

Along an invariant curve, parametrise the position by an angle ϑ that runs once around the curve. If ϑ advances by $\vartheta_{n+1} - \vartheta_n$ per kick, the *rotation number*

$$\nu = \lim_{n\to\infty} \frac{\vartheta_n - \vartheta_0}{2\pi n}$$

is the average fraction of a full turn per kick. If ν is irrational, the orbit fills the curve densely (*quasi-periodic* motion). If $\nu = r/q$ is rational, the orbit is periodic with period q. In a phase portrait, regular orbits appear as smooth closed curves.

Why small k looks integrable at $p = \pi/2$. For small k (fig. 5, $k = 0.5$), the phase portrait looks almost integrable, although the unperturbed map $R = R_y(\pi/2)$ is degenerate: $R^4 = R_y(2\pi) = I$, every orbit has period 4. So look at four periods at once. Write $F = TR$, with T the torsion, the time-1 flow of the kick Hamiltonian $h_T = \frac{k}{2}Z^2$ (Step 2 of section 3). Inserting $R^{-m}R^m = I$,

$$F^4 = TRTRTRTR = T\left(RTR^{-1}\right)\left(R^2TR^{-2}\right)\left(R^3TR^{-3}\right)R^4, \qquad R^4 = I.$$

Each factor $R^m T R^{-m}$ is again the time-1 flow of a Hamiltonian, namely $h_T \circ R^{-m}$ (h_T evaluated at $R^{-m}\boldsymbol{s}$). To see this, let $\boldsymbol{s}(t)$ follow h_T and put $\boldsymbol{u} = R^m\boldsymbol{s}$. Rotations preserve cross products, and by the chain rule $\nabla_u\big[h_T(R^{-m}\boldsymbol{u})\big] = R^m\,\nabla_s h_T$ (because $R^{-m} = (R^m)^{\mathsf{T}}$), so

$$\dot{\boldsymbol{u}} = R^m\big(\nabla_s h_T \times \boldsymbol{s}\big) = \big(R^m\nabla_s h_T\big) \times \big(R^m\boldsymbol{s}\big) = \nabla_u\big[h_T \circ R^{-m}\big] \times \boldsymbol{u}.$$

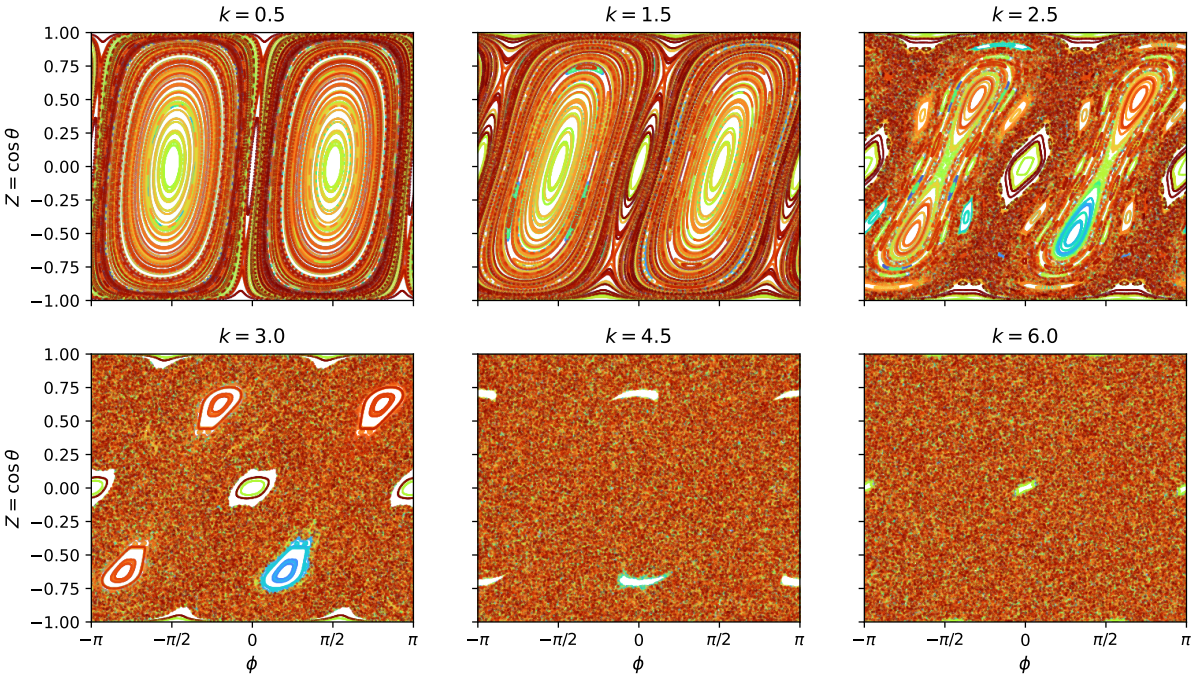


Figure 5: Phase portraits of the kicked top for $p = \pi/2$ and increasing k, in the canonical coordinates $(\phi, Z = \cos\theta)$ (equal areas in the plot are equal areas on the sphere). Each panel shows a 20×20 equal-area grid of initial conditions iterated for 500 kicks, one colour per orbit. Closed curves are regular (quasi-periodic) orbits. Uniformly speckled regions are filled by chaotic orbits.

From (11), the z component of $R^{-m}\boldsymbol{s}$ is Z, X, $-Z$, $-X$ for $m = 0, 1, 2, 3$, so the four Hamiltonians are $\frac{k}{2}Z^2$, $\frac{k}{2}X^2$, $\frac{k}{2}Z^2$, $\frac{k}{2}X^2$. Each factor is the identity plus a correction of order k, so to first order in k the product is the time-1 flow of the sum,

$$\bar{h} = \frac{k}{2}\left(Z^2 + X^2 + Z^2 + X^2\right) = k\left(X^2 + Z^2\right) = k\left(1 - Y^2\right). \tag{32}$$

$\nabla_s \bar{h} = -2kY\,\hat{\boldsymbol{y}}$, so the flow of $\bar{h}$ is a rotation about y with rate $-2kY$, and Y is conserved. The invariant curves of F^4 are therefore close to circles $Y = \sqrt{1 - Z^2}\sin\phi = \text{const}$ around the poles $\pm\hat{\boldsymbol{y}}$. In the (ϕ, Z) plane these are the closed curves around $(\pm\pi/2, 0)$ in the $k = 0.5$ panel. On the great circle $Y = 0$ (the lines $\phi = 0, \pm\pi$) the rate $-2kY$ vanishes, so higher-order terms decide what happens there. Numerically a thin chaotic layer forms along these lines.

7.2 KAM theorem and the breakup of invariant curves

What happens to invariant curves when an integrable map is perturbed? The answer is given by the Kolmogorov–Arnold–Moser (KAM) theorem. Call a number ν *sufficiently irrational* if it is badly approximated by fractions: $|\nu - r/q| \geq C/q^\tau$ for all integers r, q, for some constants $C > 0$ and $\tau \geq 2$. Stated informally:

Theorem (KAM, informal)**.** *For a sufficiently small smooth perturbation of an integrable area-preserving twist map, most invariant curves survive, only slightly deformed. The survivors are those whose rotation number ν is sufficiently irrational. The destroyed curves occupy a fraction of phase space that goes to zero with the perturbation.*

The curves with rational rotation number $\nu = r/q$ do not survive. By the Poincaré–Birkhoff theorem, such a curve breaks up into a chain of alternating elliptic and hyperbolic period-q

points. Around the elliptic ones there are new small islands. Each hyperbolic point has a *stable manifold* (the curve of points that approach it under repeated application of F) and an *unstable manifold* (the points that approach it under F^{-1}). These two curves intersect in an infinitely folded pattern (a *homoclinic tangle*), which produces a thin chaotic layer. The structure repeats on ever finer scales: islands around islands.

As k grows, the chaotic layers widen. When neighbouring resonances (island chains) grow until they overlap (Chirikov's resonance-overlap criterion), the KAM curves that separated them are destroyed and the layers merge into a connected *chaotic sea*. In a 2D map, surviving KAM curves are impenetrable barriers: a chaotic orbit can never enter an island and a regular orbit can never leave one.

7.3 Mixed and global chaos in the kicked top

Figure 5 shows the scenario for $p = \pi/2$:

- $k = 0.5$ and 1.5: almost everything is regular. Large families of closed curves surround the stable poles $\pm\hat{\boldsymbol{y}}$ ($|\operatorname{Tr} M_{\pm}| = k < 2$, (28)), with thin chaotic layers near $\phi = 0, \pm\pi$.
- $k = 2.5$: the poles have just become unstable (section 6.2). Chaos spreads from them, and the new islands of $\boldsymbol{s}_{\pm}$ and of the period-2 orbit appear, together with a mixture of smaller island chains.
- $k = 3$: a *mixed phase space*. A large chaotic sea surrounds islands around $\boldsymbol{s}_{\pm}$ (at $(0.63, -0.63)$ and its mirror image $(2.51, 0.63)$), around the period-2 orbit (at $(-0.63, 0.63)$ and $(-2.51, -0.63)$), and around the period-4 orbit (at $(0, 0)$, $(\pm\pi, 0)$ and near $Z = \pm 1$). The pairing of islands under S of (25) is clearly visible.
- $k = 4.5$ and 6: almost *global chaos*. The islands of $\boldsymbol{s}_{\pm}$ are gone ($k > \pi\sqrt{2}$) and only tiny islands remain (e.g. the period-4 island at $(0, 0)$ for $k = 6$, where $|\operatorname{Tr} M^{(4)}| < 2$ by (31)).

Figure 6 shows the difference between the two kinds of motion for $k = 3$. A single orbit started inside the island around $\boldsymbol{s}_{+}$ stays on a closed curve. A single orbit started next to the unstable pole $+\hat{\boldsymbol{y}}$ wanders through the whole chaotic sea but never enters any island. Figure 7 shows the $k = 3$ portrait on the sphere itself.

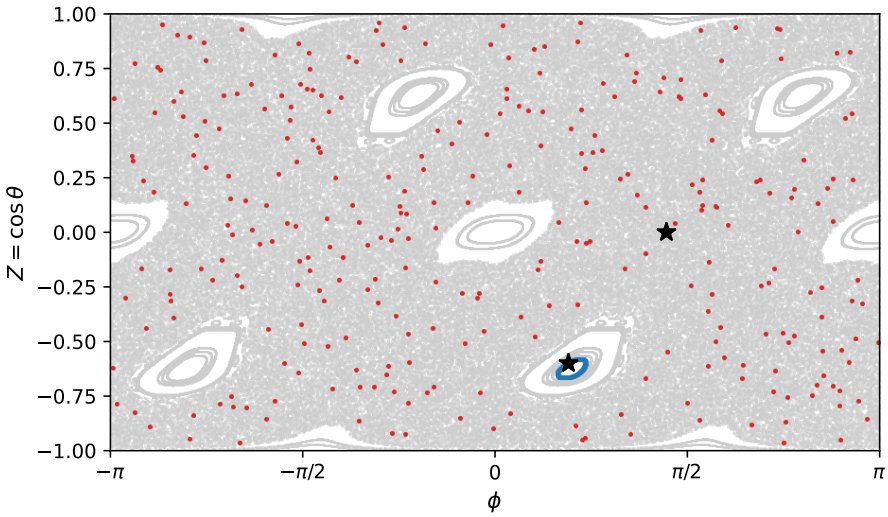


Figure 6: Two single orbits for $p = \pi/2$, $k = 3$, 300 kicks each, on top of the full portrait (grey). Blue: started at $(\phi_0, Z_0) = (0.6, -0.6)$, inside the island of $\boldsymbol{s}_{+}$. Red: started at $(1.4, 0)$, near the hyperbolic pole $+\hat{\boldsymbol{y}}$. Stars mark the starting points.

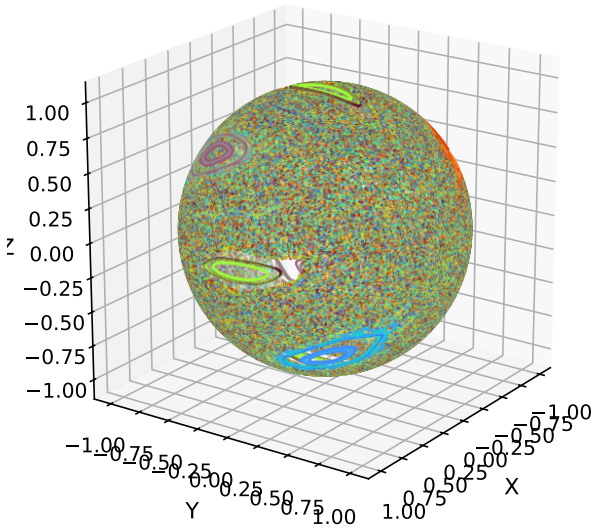


Figure 7: The $k = 3$ phase portrait of fig. 5 drawn on the unit sphere (X, Y, Z).

8 Lyapunov exponents

Chaos means *sensitive dependence on initial conditions*. Two orbits starting a small distance d_0 apart separate exponentially, $d_n \sim d_0\, e^{\Lambda n}$. The rate Λ is the (maximal) *Lyapunov exponent*.

Definition. Let $\boldsymbol{s}_0, \boldsymbol{s}_1, \ldots$ be an orbit and $\delta\boldsymbol{s}_0$ a small displacement of its starting point. By the linearisation of section 6, $\delta\boldsymbol{s}_1 = DF(\boldsymbol{s}_0)\,\delta\boldsymbol{s}_0$, $\delta\boldsymbol{s}_2 = DF(\boldsymbol{s}_1)\,\delta\boldsymbol{s}_1$, and so on, so $\delta\boldsymbol{s}_n = M_n\,\delta\boldsymbol{s}_0$ with the *tangent map*

$$M_n = DF(\boldsymbol{s}_{n-1}) \cdots DF(\boldsymbol{s}_1)\, DF(\boldsymbol{s}_0).$$

With $\|\boldsymbol{v}\|$ the length of a vector,

$$\Lambda = \lim_{n\to\infty} \frac{1}{n} \ln \frac{\|M_n\,\delta\boldsymbol{s}_0\|}{\|\delta\boldsymbol{s}_0\|}, \tag{33}$$

which for almost every $\delta\boldsymbol{s}_0$ does not depend on the direction of $\delta\boldsymbol{s}_0$. For an area-preserving 2D map $\det M_n = 1$: stretching by $e^{\Lambda n}$ in one direction is compensated by shrinking by $e^{-\Lambda n}$ in another, so the two exponents are $\pm\Lambda$. Typical values:

- regular (quasi-periodic) orbits: $\Lambda = 0$. Nearby orbits on neighbouring invariant curves separate only linearly, $d_n \propto n$, because their rotation numbers differ slightly, and $\frac{1}{n}\ln n \to 0$;
- a hyperbolic fixed point with eigenvalue λ_u: there $M_n = M^n$, so $\|M_n\delta\boldsymbol{s}_0\| \sim |\lambda_u|^n$ and $\Lambda = \ln|\lambda_u|$. For the pole $+\hat{\boldsymbol{y}}$ at $k = 3$, (28) gives $\operatorname{Tr} M = 3$, so $\lambda^2 - 3\lambda + 1 = 0$, $\lambda_u = (3+\sqrt{5})/2 \approx 2.62$ and $\Lambda \approx 0.96$;
- chaotic orbits: $\Lambda > 0$. The value is an average over the whole chaotic sea, which contains regions of very different local stretching.

Figure 8 shows this directly. Pairs of orbits are started with θ_0 differing by 10^{-8}. In the island the distance stays at $\sim 10^{-8}$. In the chaotic sea it grows exponentially until it saturates at the size of the sphere (at most 2, the diameter). Taking the logarithm of $d_n \sim d_0 e^{\Lambda n}$ gives $\ln d_n \approx \ln d_0 + \Lambda n$, a straight line of slope Λ. A straight-line fit of $\ln d_n$ for $5 \le n \le 40$ gives a

slope of about 0.47 per kick. This is a rough, finite-time estimate from a single pair. A reliable value of Λ needs (33) evaluated over long times, with $\delta\boldsymbol{s}$ rescaled to unit length every few kicks so that it never leaves the linear regime (the method of Benettin *et al.* [7]). The code computes Λ in exactly this way (section 8.1).

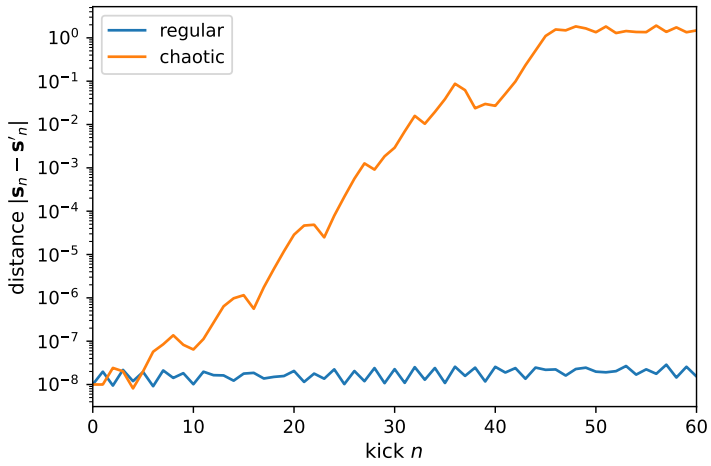


Figure 8: Distance between two orbits whose initial θ_0 differ by 10^{-8}, for $p = \pi/2$, $k = 3$. Blue: both inside the island of $\boldsymbol{s}_+$. Orange: both in the chaotic sea. The straight segment on the logarithmic axis is exponential separation.

8.1 Computing Λ: the tangent map and the Benettin algorithm

The tangent map for any p. The definition (33) needs $DF(\boldsymbol{s})\,\delta\boldsymbol{s}$, the image of a small displacement under one kick. Write the map (20) as $F(\boldsymbol{s}) = R_z\big(kZ'(\boldsymbol{s})\big)\,\boldsymbol{s}'(\boldsymbol{s})$ with $\boldsymbol{s}' = R_y(p)\,\boldsymbol{s}$ and $Z' = (\boldsymbol{s}')_z$, and vary $\boldsymbol{s} \to \boldsymbol{s} + \delta\boldsymbol{s}$.

1. The rotation is linear: $\delta\boldsymbol{s}' = R_y(p)\,\delta\boldsymbol{s}$, and $\delta Z' = (\delta\boldsymbol{s}')_z$.

2. The torsion depends on $\boldsymbol{s}'$ twice, through the vector it rotates and through its angle kZ'. By the product rule,

$$\delta\big[R_z(kZ')\,\boldsymbol{s}'\big] = R_z(kZ')\,\delta\boldsymbol{s}' + \left.\frac{dR_z}{d\alpha}\right|_{\alpha=kZ'} \boldsymbol{s}'\,k\,\delta Z'.$$

3. With $\frac{d}{d\alpha}R_z(\alpha)\boldsymbol{v} = \hat{\boldsymbol{z}} \times R_z(\alpha)\boldsymbol{v}$ (section 1.4) and $R_z(kZ')\boldsymbol{s}' = \boldsymbol{s}_{n+1}$, the last term is $k\,\delta Z'\,\hat{\boldsymbol{z}} \times \boldsymbol{s}_{n+1}$.

Hence

$$\boxed{\delta\boldsymbol{s}_{n+1} = DF(\boldsymbol{s}_n)\,\delta\boldsymbol{s}_n = R_z(kZ')\,\delta\boldsymbol{s}' + k\,\delta Z'\,\hat{\boldsymbol{z}} \times \boldsymbol{s}_{n+1}, \qquad \delta\boldsymbol{s}' = R_y(p)\,\delta\boldsymbol{s}_n.} \tag{34}$$

(For $p = \pi/2$ and the points of the period-4 orbit this reduces to the DT used in section 6.3.) The image is again tangent to the sphere. If $\boldsymbol{s}_n \cdot \delta\boldsymbol{s}_n = 0$, then, because rotations preserve dot products and $\hat{\boldsymbol{z}} \times \boldsymbol{s}_{n+1}$ is perpendicular to $\boldsymbol{s}_{n+1}$,

$$\boldsymbol{s}_{n+1} \cdot \delta\boldsymbol{s}_{n+1} = \big(R_z\boldsymbol{s}'\big) \cdot \big(R_z\delta\boldsymbol{s}'\big) = \boldsymbol{s}' \cdot \delta\boldsymbol{s}' = \boldsymbol{s}_n \cdot \delta\boldsymbol{s}_n = 0.$$

The algorithm. $\delta\boldsymbol{s}_n = M_n\delta\boldsymbol{s}_0$ grows like $e^{\Lambda n}$, and for $\Lambda \approx 0.4$ it overflows double precision ($\approx 10^{308}$) after about $\ln(10^{308})/0.4 \approx 1800$ kicks. Since (34) is linear in $\delta\boldsymbol{s}$, the vector can be rescaled at any time without changing its direction. Benettin's algorithm does exactly this:

1. start from a unit tangent vector $\delta\boldsymbol{s}_0$, $\boldsymbol{s}_0 \cdot \delta\boldsymbol{s}_0 = 0$;
2. apply (34) and (20) together, kick by kick;
3. every m kicks, measure the length $d_j = \|\delta\boldsymbol{s}\|$ and rescale, $\delta\boldsymbol{s} \to \delta\boldsymbol{s}/d_j$;
4. after N kicks, $\Lambda_N = \frac{1}{N}\sum_j \ln d_j$.

Because every rescaling only divides by the length just measured, $\prod_j d_j = \|M_N\,\delta\boldsymbol{s}_0\|$, so Λ_N is exactly (33) evaluated at the finite time N, without overflow. Λ_N converges to Λ as $N \to \infty$. At finite N there are two kinds of error. The first part of the evolution, before $\delta\boldsymbol{s}$ has turned into the most unstable direction, contributes a term of order $1/N$. On regular orbits $\|\delta\boldsymbol{s}\|$ grows like n, so $\Lambda_N \approx \ln N/N$, small but not zero.

Result for $k = 3$. Figure 9 shows Λ_N ($N = 2000$) for an 80×80 grid of initial conditions. The islands of fig. 5 appear as regions with $\Lambda_N \approx 0$, and the chaotic sea has a nearly uniform value $\Lambda \approx 0.33$, as it must, since a single chaotic orbit visits the whole sea. (The larger slope ≈ 0.47 of fig. 8 is a local rate over the first 40 kicks near the pole $+\hat{\boldsymbol{y}}$, where the stretching $\ln\lambda_u \approx 0.96$ is stronger than the average.)

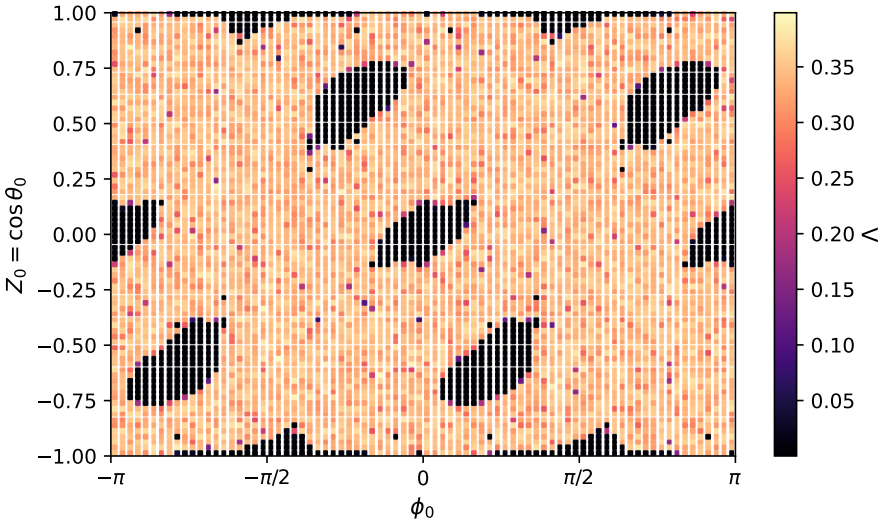


Figure 9: Finite-time Lyapunov exponent Λ_N ($N = 2000$ kicks, Benettin algorithm with rescaling every kick) of each initial condition of an 80×80 equal-area grid, $p = \pi/2$, $k = 3$. Dark: regular islands ($\Lambda_N \approx 0$). Bright: chaotic sea ($\Lambda \approx 0.33$).

9 Outlook: the quantum kicked top

The kicked top was introduced by Haake, Kuś and Scharf [1] as a model for *quantum* chaos. In quantum mechanics the components of $\boldsymbol{J}$ become operators $\hat{J}_x, \hat{J}_y, \hat{J}_z$ (a hat marks an operator in this section) with $[\hat{J}_a, \hat{J}_b] = i\,\epsilon_{abc}\hat{J}_c$ in units $\hbar = 1$. They act on a $(2j+1)$-dimensional space of state vectors $|\psi\rangle$, and $\hat{\boldsymbol{J}}^2 = j(j+1)$. The state obeys the Schrödinger equation $i\,\frac{d}{dt}|\psi\rangle = \hat{H}(t)|\psi\rangle$ with $\hat{H}$ given by (12). Between kicks $\hat{H} = p\hat{J}_y$ is constant, and the evolution over the unit time is $\exp(-ip\hat{J}_y)$. Across a kick the δ term dominates, as in Step 2 of section 3, and gives $\exp\!\big(-i\frac{k}{2j}\hat{J}_z^2\big)$. One period is therefore

$$\hat{U} = \exp\Big(-i\frac{k}{2j}\hat{J}_z^2\Big)\exp\big(-ip\hat{J}_y\big), \qquad |\psi_{n+1}\rangle = \hat{U}|\psi_n\rangle, \tag{35}$$

the quantum counterpart of (20): first the rotation, then the kick. In terms of $\hat{\boldsymbol{s}} = \hat{\boldsymbol{J}}/j$ the commutator is $[\hat{s}_x, \hat{s}_y] = [\hat{J}_x, \hat{J}_y]/j^2 = \frac{i}{j}\hat{s}_z$, so $1/j$ acts as an effective Planck constant and the classical map of these notes is the limit $j \to \infty$. A spin coherent state, a minimum-uncertainty wave packet of width of order $1/\sqrt{j}$ centred at a point $\boldsymbol{s}$ of the sphere, follows the classical orbit until it spreads. On a chaotic orbit the width grows like $e^{\Lambda n}/\sqrt{j}$ and becomes of order 1 after $n \sim \ln j/(2\Lambda)$ kicks. On regular orbits the spreading is much slower.

The classical phase-space structure leaves fingerprints in the quantum system. Examples include the statistics of the eigenphases of $\hat{U}$ (Poisson-like for regular, random-matrix-like for chaotic dynamics), the way eigenstates concentrate on classical islands or spread over the chaotic sea, and the entanglement generated when the spin is made of many qubits. The book by Haake [2] is the standard reference. The kicked top has been realised experimentally with the hyperfine spin of cesium atoms [8] and with superconducting qubits [9].

Part II
Numerics and code

10 Numerical method

Exact map, no integrator. The equations of motion (15) are solved exactly over one period (section 3), so the code never integrates a differential equation. It applies the closed-form map (20): two rotations per kick, 4 trigonometric evaluations and a few multiplications. There is no time-step error, and the only error is floating-point round-off. Since both steps are exact rotations, the norm $|\boldsymbol{s}| = 1$ and the area preservation of section 3.1 hold up to round-off. In practice $\big||\boldsymbol{s}_n| - 1\big| \lesssim 3 \times 10^{-15}$ after 500 kicks.

Round-off on chaotic orbits. On a chaotic orbit any error, including a round-off error of 10^{-16}, grows like $e^{\Lambda n}$. With $\Lambda \approx 0.5$ a computed chaotic trajectory differs from the exact trajectory with the same initial condition after roughly $\ln(10^{16})/0.5 \approx 70$ kicks. This does not make the phase portraits wrong. The computed orbit is still an orbit of the map to round-off accuracy at every step, and (by shadowing arguments) it stays close to some true orbit, so it explores the same chaotic sea. Individual chaotic trajectories beyond that time should be read statistically, not pointwise.

Initial conditions. Three modes are available.

- `single`: one point given by (θ_0, ϕ_0) through (13).
- `grid`: an $n_\theta \times n_\phi$ grid of cell centres, uniform in $Z = \cos\theta \in [-1, 1]$ and in $\phi \in [-\pi, \pi]$. By (9) every grid cell has the same phase-space area, so the grid samples all regions of phase space with equal weight. A grid uniform in θ would oversample the poles.
- `random`: points with Z and ϕ uniformly distributed, which by (9) is the uniform distribution on the sphere.

Recording. After `n_transient` discarded kicks, the state is stored at kicks $n = 0, m, 2m, \cdots \leq$ `n_kicks`, where $m =$ `save_every`. The initial point is stored as $n = 0$ and the map is applied once per kick, so stored points are always "just after a kick", as in section 3.

Parallelisation. Different initial conditions are independent, so the loop over them is parallelised with OpenMP. The default configuration (20×20 grid, 500 kicks) runs in under a second.

11 Code structure

The physics is in Fortran (`src/`); the Fortran code writes plot-ready files, and Python only reads and plots them. Figure 10 shows the data flow and table 1 maps each module to the equations it implements. In the source, the header comment of every module and function cites the equations it implements by number and by LaTeX label, e.g. `! ... eq. (20) [eq:map]`, so the reference can be found again with `grep` if the numbering of these notes changes.

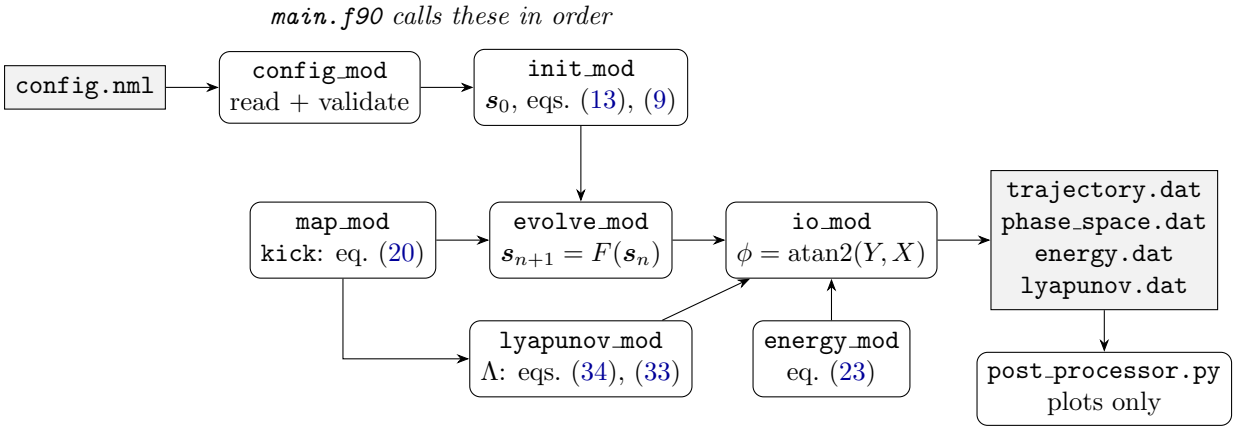


Figure 10: Data flow of the program. `main.f90` reads the configuration, builds the initial conditions, evolves them and writes the output. If `&lyapunov` is enabled, it also computes Λ for every initial condition.

File	Role	Physics
`kinds_mod.f90`	working precision `wp` (double), constants π, 2π	–
`config_mod.f90`	type `config_t`; reads the namelist groups, checks values	parameters p, k of (12)
`init_mod.f90`	`initial_conditions`: `single` / `grid` / `random`; `spherical_to_cartesian`	(13); equal-area grid from (9)
`map_mod.f90`	pure functions `kick(s, p, k)`: one period; `kick_tangent(s, ds, p, k)`: its tangent map; `rotate_y`, `rotate_z`: the two rotations	(17), (19), (20), (34)
`evolve_mod.f90`	iterates `kick` for all initial conditions (OpenMP), transient and `save_every`	$s_n = F^n(s_0)$
`lyapunov_mod.f90`	`lyapunov`: Benettin algorithm for every initial condition (OpenMP)	(33), section 8.1
`energy_mod.f90`	pure function `energy(s, p, k)` $\to [pY,\ \frac{k}{2}Z^2,\ \text{total}]$	(23)
`io_mod.f90`	writes the ASCII output files; computes $\phi = \mathrm{atan2}(Y, X) \in (-\pi, \pi]$	coordinates of section 1.3
`main.f90`	driver: `read_config` $\to$ `initial_conditions` $\to$ `evolve` $\to$ `write_results` ($\to$ `lyapunov` $\to$ `write_lyapunov`)	–
`post_processor.py` (in `scripts/`)	class `PostProcessor`: phase-space plot (ϕ, Z), 3D plot on the sphere, map of Λ	–

Table 1: Files of the code and the equations they implement.

The map. `kick` in `map_mod.f90` applies `rotate_y`, the rotation (17), and then `rotate_z` with the angle kZ', the torsion (19). `kick_tangent` is the tangent map (34); its last line is $R_z(kZ')\,\delta\boldsymbol{s}' + k\,\delta Z'\,\hat{\boldsymbol{z}} \times \boldsymbol{s}_{n+1}$, with $\hat{\boldsymbol{z}} \times \boldsymbol{s} = (-Y, X, 0)$:

```
  ! One period of the kicked top, s_{n+1} = F(s_n), eq. (20) [eq:map]:
  ! free precession about y by the angle p, eq. (17) [eq:rotation-y],
  ! followed by the torsion about z by the angle k*Z', eq. (19) [eq:torsion].
  pure function kick(s, p, k) result(s_new)
    real(wp), intent(in) :: s(3)
    real(wp), intent(in) :: p
    real(wp), intent(in) :: k
    real(wp) :: s_new(3)
    real(wp) :: s1(3)

    s1 = rotate_y(s, p)
    s_new = rotate_z(s1, k*s1(3))
  end function kick

  ! Tangent map: the image DF(s)*ds of a small displacement ds of s under one
  ! kick, eq. (34) [eq:tangent]. The rotation is linear; the torsion
  ! R_z(k*Z')s' also turns with the change k*dZ' of its angle, which adds
  ! k*dZ' * (z_hat x s_{n+1}). If ds is tangent to the sphere at s, the result
  ! is tangent at F(s).
  pure function kick_tangent(s, ds, p, k) result(ds_new)
    real(wp), intent(in) :: s(3)
    real(wp), intent(in) :: ds(3)
    real(wp), intent(in) :: p
    real(wp), intent(in) :: k
    real(wp) :: ds_new(3)
    real(wp) :: s1(3), ds1(3), s2(3)

    s1 = rotate_y(s, p)
    ds1 = rotate_y(ds, p)
    s2 = rotate_z(s1, k*s1(3))
    ds_new = rotate_z(ds1, k*s1(3)) + k*ds1(3)*[-s2(2), s2(1), 0.0_wp]
  end function kick_tangent
```

The equal-area grid. `grid_points` in `init_mod.f90` places cell centres uniformly in Z and ϕ, as required by (9):

```
    do i = 1, n_theta
      z = -1.0_wp + (real(i, wp) - 0.5_wp)*2.0_wp/real(n_theta, wp)
      do j = 1, n_phi
        phi = -pi + (real(j, wp) - 0.5_wp)*two_pi/real(n_phi, wp)
        ic = ic + 1
        s0(:, ic) = spherical_to_cartesian(acos(z), phi)
      end do
    end do
```

11.1 Parameters

All parameters are read from a namelist file (default `config.nml`; another file can be passed as the first command-line argument). Missing groups or variables keep their defaults.

Group	Variable	Meaning
`&system`	`p`, `k`	rotation angle p and torsion strength k in (12)
`&simulation`	`n_kicks`	number of kicks recorded after the transient
	`n_transient`	kicks discarded before recording
	`save_every`	store every `save_every`-th kick
`&initial`	`mode`	`'single'`, `'grid'` or `'random'`
	`theta0`, `phi0`	initial point for `single` (radians)
	`n_theta`, `n_phi`	grid size in $Z = \cos\theta$ and in ϕ
	`n_random`, `seed`	number of random points and seed
`&output`	`out_dir`	output directory
`&lyapunov`	`enabled`	compute Λ_N ($N =$ `n_kicks`) for every initial condition
	`n_renorm`	rescale $\delta\boldsymbol{s}$ every `n_renorm` kicks (m in section 8.1)

11.2 Output files

All files are ASCII, start with a header line `# p = ..., k = ...` and a column line, and contain one block per initial condition separated by a blank line. The first three files have the same row order, so row r of each refers to the same point $\boldsymbol{s}_n$. `lyapunov.dat` (written only if `&lyapunov` is enabled) has one line per initial condition.

File	Columns
`trajectory.dat`	orbit index `ic`, kick `n`, X, Y, Z
`phase_space.dat`	`ic`, `n`, ϕ, Z (canonical coordinates, section 1.3)
`energy.dat`	pY, $\frac{k}{2}Z^2$, total (eq. (23))
`lyapunov.dat`	`ic`, ϕ_0, Z_0 (initial point), Λ_N (section 8.1)

11.3 Building and running

```
make # build/kicked_top
./build/kicked_top config.nml # or: make run
python scripts/post_processor.py # output/phase_space.png
python scripts/post_processor.py --sphere --show
python scripts/post_processor.py --lyapunov # output/lyapunov.png
make figures # recompute the figures of these notes
make docs # compile docs/kicked_top_notes.tex
```

The figures of these notes are reproducible. `make figures` runs the program on every namelist in `docs/figures/configs/`, writing the output to `docs/figures/data/`. Then the script `plot_figures.py` in `docs/figures/` draws the PDF figures from those files.

11.4 Verification

The following checks were carried out on the code:

- $k = 0$, $p = \pi/2$: the map is $R_y(\pi/2)$, so every orbit must return to its starting point after 4 kicks. Observed error: 5×10^{-16}.
- $\big||\boldsymbol{s}_n| - 1\big| \le 3.4 \times 10^{-15}$ for the default run (400 orbits, 500 kicks, $k = 3$).
- The three columns of `energy.dat` agree with pY, $\frac{k}{2}Z^2$ and their sum recomputed from `trajectory.dat` to 10^{-15}.
- The fixed points, traces and symmetry of sections 5 and 6 were checked against the map numerically, and they agree with the islands in fig. 5.

- `kick_tangent` agrees with a centred finite difference of `kick` to 10^{-10} (relative, for $\|\delta\boldsymbol{s}\| \approx 10^{-7}$).
- Λ at the hyperbolic pole $+\hat{\boldsymbol{y}}$, $k = 3$: the $1/N$ term of section 8.1 cancels in $(30\Lambda_{30} - 20\Lambda_{20})/10 = 0.96242365$, which agrees with the exact $\ln\frac{3+\sqrt{5}}{2} = 0.96242365$ (section 8) to 3×10^{-9}. (Longer runs leave the pole, because round-off moves the orbit off it.)
- $k = 0$: $|\Lambda_N| < 10^{-17}$ for every initial condition, as it must be for a pure rotation.

11.5 Extending the code

New functionality follows the same pattern: one module with its own namelist group and an on/off switch, called from `main.f90`. The Lyapunov exponent is an example:

1. `map_mod` provides the tangent map `kick_tangent` (34) next to `kick`, built from the same `rotate_y` and `rotate_z`;
2. `lyapunov_mod.f90` evolves $(\boldsymbol{s}, \delta\boldsymbol{s})$ together and implements the algorithm of section 8.1;
3. the namelist group `&lyapunov` switches it on, `io_mod` writes `lyapunov.dat`, and the method `plot_lyapunov` of `PostProcessor` plots it.

Other variants (a different kick axis, several kicks per period, dissipation) only replace `kick` and `kick_tangent`. The rest of the program does not depend on the form of the map.

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